

Fundamental Principles

FE Environmental — Day 10

Study set • 2026-06-26 • 18 questions • every numeric answer recomputed in Python

Handbook fidelity. All formulas confirmed against NCEES FE Reference Handbook v10.6. Population projection equations (p. 334): Linear $P_t = P_0 + kt$, Geometric $P_t = P_0 e^{kt}$, Percent $P_t = P_0(1+k)^n$, Ratio-Correlation $P_2/P_2R = P_1/P_1R$, Decreasing-Rate-of-Increase $P_t = P_0 + (S - P_0)(1 - e^{-k(t-t_0)})$ — all five models verbatim. Steady-State Reactor Parameters table (p. 330–331): CMFR and PFR first-order, zero-order, and second-order retention times and effluent concentrations verbatim. Engineering stress $= F/A_0$, true stress $T = F/A$, Hooke's Law $= E\varepsilon$ (p. 121). Thermal expansion coefficient $\varepsilon = \alpha T$ (p. 127 Materials Science). Thermal deformations $t = \alpha L(T - T_0)$ (p. 131 Mechanics of Materials). Fire hydrant rated capacity $QR = QF \times (HR/HF)^{0.54}$ and discharge $Q = 29.8 D^2 C_d P^{0.5}$ (p. 304–305, Civil Engineering, NFPA 291). NOT in Handbook v10.6 (supplemental): the term 'logistic growth' (Handbook calls it Decreasing-Rate-of-Increase Growth); 'arithmetic growth' (Handbook calls it Linear Projection = Algebraic Projection); 'geometric' appears only in the synonym list 'Log Growth = Exponential Growth = Geometric Growth' — all questions use Handbook nomenclature. The CMFR 2nd-order effluent formula $C_t = (\sqrt{4kC_0\theta + 1} - 1)/(2k\theta)$ is derived by solving the CMFR steady-state equation; the Handbook table gives the θ formula, from which C_t is algebraically equivalent — tagged as handbook. True stress formula $T = \times(1 + \varepsilon)$ for constant-volume plastic deformation is derivable from Handbook definitions ($T = F/A$, $\varepsilon = F/A_0$) but the combined form is not explicitly printed; Q14 gives both areas explicitly so students use $T = F/A$ directly (handbook). For USCS Q13: $E = 29 \times 10^6$ psi is a standard steel value not tabulated in Handbook — provided in the stem (tagged in-stem). Ratio-correlation: $k = P_2R/P_1R$ (the constant) is rearranged from the Handbook form; the formula itself is in Handbook (tagged handbook).

Q1. [D10-Q01 • MCQ • Population projection — linear (algebraic) model]

A small city had a population of 25,000 in 2010. Using the linear (algebraic) projection model, the population in 2020 is most nearly:

Linear Projection: $P_t = P_0 + kt$

The growth rate $k = 800$ persons/year (determined from historical data).

Relevant: $P_t = P_0 + k \Delta t$

- (A) 30,000
- (B) 31,000
- (C) **33,000 ✓**
- (D) 35,000

Answer: (C)

Using the linear projection formula: $P_t = P_0 + k\Delta t = 25,000 + 800 \times 10 = 25,000 + 8,000 = 33,000$.

Option (A) 30,000: uses $k = 500$ persons/yr (too low). Option (B) 31,000: uses $k = 600$ persons/yr. Option (D) 35,000: uses $k = 1,000$ persons/yr (too high). The linear model assumes a constant annual increment — no compounding.

Handbook: Environmental Engineering — Population Projection Equations, Linear Projection = Algebraic Projection: $P_t = P_0 + kt$ (p. 334) • Handbook p. 334 — Population Projection Equations

Q2. [D10-Q02 • MCQ • Population projection — geometric (log/exponential) model]

Census data for a municipality show a population of 50,000 in 2010 and 65,000 in 2020. Assuming geometric (log) growth:

Log Growth = Exponential Growth = Geometric Growth: $P_t = P_0 e^{(kt)}$

The projected population in 2025 is most nearly:

Relevant: $P_t = P_0 e^{k\Delta t}$ $k = \frac{\ln(P_t/P_0)}{\Delta t}$

- (A) 70,000
- (B) 72,000
- (C) **74,000** ✓
- (D) 76,000

Answer: (C)

Step 1 — find k from the two census points (2010→2020, t = 10 yr):

$$k = \frac{\ln(65,000/50,000)}{10} = \frac{\ln(1.30)}{10} = \frac{0.2624}{10} = 0.02624 \text{ yr}^{-1}$$

Step 2 — project from 2020 to 2025 (t = 5 yr from 2020 base):

$$P_{2025} = 65,000 \times e^{0.02624 \times 5} = 65,000 \times e^{0.1312} = 65,000 \times 1.1401 = 74,107$$

Option (A) 70,000: misapplies k, uses only additive estimate. Option (B) 72,000: underestimates k. Option (D) 76,000: overestimates. The geometric model compounds continuously — larger populations grow faster each year.

Handbook: Environmental Engineering — Population Projection Equations, Log Growth = Exponential Growth = Geometric Growth: $P_t = P_0 e^{(kt)}$ (p. 334) • Handbook p. 334 — Population Projection Equations

Q3. [D10-Q03 • MCQ • Population projection — percent (discrete) growth model]

A growing town has a current population of 80,000. It is expected to grow at 2.5% per year. Using the percent growth model, the population after 8 years is most nearly:

Percent Growth: $P_t = P_0(1 + k)^n$

Relevant: $P_t = P_0(1 + k)^n$

- (A) 92,000
- (B) 97,000
- (C) **97,500 ✓**
- (D) 102,000

Answer: (C)

$$P_t = 80,000 \times (1 + 0.025)^8 = 80,000 \times (1.025)^8.$$

$$(1.025)^8: (1.025)^2 = 1.0506; (1.025)^4 = 1.1038; (1.025)^8 = 1.2184.$$

$$P_t = 80,000 \times 1.2184 = 97,472 \approx 97,500.$$

Option (A) 92,000: uses $n = 5$ instead of 8. Option (B) 97,000: rounds k incorrectly. Option (D) 102,000: misapplies $k = 3\%$. The percent growth model is discrete (annual compounding), unlike the geometric model which compounds continuously.

Handbook: Environmental Engineering — Population Projection Equations, Percent Growth: $P_t = P_0(1+k)^n$ (p. 334) • Handbook p. 334 — Population Projection Equations

Q4. [D10-Q04 • MCQ • Population projection — ratio-and-correlation method]

At the last census a city had a population of 40,000 and its surrounding metropolitan region had 500,000. In the new forecast, the region is projected to reach 620,000. Using the ratio-and-correlation method, the projected city population is most nearly:

Ratio and Correlation Growth: $P_2/P_{2R} = P_1/P_{1R}$

Relevant: $\frac{P_2}{P_{2R}} = \frac{P_1}{P_{1R}}$

- (A) 44,800
- (B) 46,500
- (C) **49,600 ✓**
- (D) 52,000

Answer: (C)

The ratio-correlation method assumes the city's share of the regional population stays constant:

$$P_2 = P_{2R} \times \frac{P_1}{P_{1R}} = 620,000 \times \frac{40,000}{500,000} = 620,000 \times 0.080 = 49,600$$

Option (A) 44,800: divides by 620 instead of multiplying by the ratio. Option (B) 46,500: uses ratio of 40/500 but applies to wrong base. Option (D) 52,000: inflates the ratio. The method scales the city in proportion to its region — useful when regional forecasts are more reliable than local ones.

Handbook: Environmental Engineering — Population Projection Equations, Ratio and Correlation Growth: $P_2/P_{2R} = P_1/P_{1R}$ (p. 334) • Handbook p. 334 — Population Projection Equations

Q5. [D10-Q05 • MCQ • Population projection — decreasing-rate-of-increase (saturation) model]

A city currently has a population of 30,000. Analysis of historical data indicates a saturation population $S = 80,000$ and a growth rate constant $k = 0.10 \text{ yr}^{-1}$. Using the decreasing-rate-of-increase model, the population 15 years from now is most nearly:

Decreasing-Rate-of-Increase Growth: $P_t = P_0 + (S - P_0)(1 - e^{-k(t-t_0)})$

Relevant: $P_t = P_0 + (S - P_0)(1 - e^{-k(t-t_0)})$

- (A) 58,000
- (B) 64,000
- (C) **68,800** ✓
- (D) 76,000

Answer: (C)

$$P_t = 30,000 + (80,000 - 30,000)(1 - e^{-0.10 \times 15}) = 30,000 + 50,000(1 - e^{-1.5}) = 30,000 + 50,000(1 - 0.2231) = 30,000 + 50,000 \times 0.7769 = 30,000 + 38,843 = 68,843 \approx 68,800.$$

Option (A) 58,000: uses $t = 10 \text{ yr}$ instead of 15. Option (B) 64,000: uses $k = 0.07$. Option (D) 76,000: assumes nearly complete saturation (overestimates). This model captures the idea that growth slows as the population approaches its carrying capacity S .

Handbook: Environmental Engineering — Population Projection Equations, Decreasing-Rate-of-Increase Growth: $P_t = P_0 + (S - P_0)(1 - e^{-k(t-t_0)})$ (p. 334) • Handbook p. 334 — Population Projection Equations

Q6. [D10-Q06 • MCQ • CMFR — first-order reaction, retention time]

A completely mixed flow reactor (CMFR) is used to treat an influent with concentration $C_0 = 200 \text{ mg/L}$. The treatment target is an effluent concentration $C_t = 20 \text{ mg/L}$. The first-order rate constant is $k = 0.5 \text{ d}^{-1}$. The required mean retention time θ is most nearly:

For a CMFR with first-order decay ($r = -kC$): $\theta = (C_0/C_t - 1)/k$

Relevant: $\theta = \frac{C_0/C_t - 1}{k}$

- (A) 4.6 d
- (B) 9.0 d
- (C) **18.0 d** ✓
- (D) 36.0 d

Answer: (C)

$$\theta = \frac{C_0/C_t - 1}{k} = \frac{200/20 - 1}{0.5} = \frac{10 - 1}{0.5} = \frac{9}{0.5} = 18.0 \text{ d}.$$

Option (A) 4.6 d: this is the PFR retention time for the same removal (see Q07) — wrong reactor model applied. Option (B) 9.0 d: forgets to divide by k . Option (D) 36.0 d: doubles the answer (arithmetic error on the ratio). A CMFR is always less efficient than a PFR for the same kinetics because it operates at the effluent (lowest) concentration throughout.

Handbook: Environmental Engineering — Steady-State Reactor Parameters table, Ideal CMFR, First-Order: $\theta = (C_0/C_t - 1)/k$ (p. 330–331) • Handbook p. 330–331 — Steady-State Reactor Parameters (Constant Density Systems)

Q7. [D10-Q07 • MCQ • Plug-flow reactor (PFR) — first-order reaction, retention time]

A plug-flow reactor (PFR) must achieve 90% removal of a contaminant with $C_0 = 200$ mg/L and $C_t = 20$ mg/L. The first-order rate constant $k = 0.5$ d⁻¹. The required retention time θ is most nearly:

For an ideal PFR with first-order decay: $\theta = \ln(C_0/C_t)/k$

Relevant: $\theta = \frac{\ln(C_0/C_t)}{k}$

- (A) 2.3 d
- (B) **4.6 d** ✓
- (C) 9.0 d
- (D) 18.0 d

Answer: (B)

$$\theta = \frac{\ln(200/20)}{0.5} = \frac{\ln(10)}{0.5} = \frac{2.303}{0.5} = 4.61 \text{ d} \approx 4.6 \text{ d.}$$

Option (A) 2.3 d: $\ln(10)/1.0$ — forgets to divide by k or uses $k = 1$ d⁻¹. Option (C) 9.0 d: applies CMFR numerator formula ($C_0/C_t - 1 = 9$) to PFR — wrong model. Option (D) 18.0 d: CMFR result. Comparison: the PFR achieves 90% removal in 4.6 d vs the CMFR's 18.0 d — a 3.9× advantage because the PFR maintains high driving-force concentration along the plug length.

Handbook: Environmental Engineering — Steady-State Reactor Parameters table, Ideal Plug Flow, First-Order: $\theta = \ln(C_0/C_t)/k$ (p. 330–331) • Handbook p. 330–331 — Steady-State Reactor Parameters (Constant Density Systems)

Q8. [D10-Q08 • MCQ • CMFR — zero-order reaction, retention time]

A CMFR is operated with a zero-order reaction rate $k = 10$ mg/(L·d). The influent concentration is $C_0 = 100$ mg/L and the target effluent is $C_t = 40$ mg/L. The required mean retention time θ is most nearly:

For a CMFR (and ideal batch and PFR) with zero-order decay ($r = -k$): $\theta = (C_0 - C_t)/k$

Relevant: $\theta = \frac{C_0 - C_t}{k}$

- (A) 4.0 d
- (B) **6.0 d** ✓
- (C) 10.0 d
- (D) 14.0 d

Answer: (B)

$$\theta = \frac{C_0 - C_t}{k} = \frac{100 - 40}{10} = \frac{60}{10} = 6.0 \text{ d.}$$

Option (A) 4.0 d: uses $C_0 - C_t = 40$ instead of 60 (confuses C_t and the reduction). Option (C) 10.0 d: uses $C_0/k = 100/10$ (ignores that we only need to reduce to 40, not to zero). Option (D) 14.0 d: arithmetic error. Note: for zero-order kinetics the CMFR, PFR, and batch all have the same retention time — unlike first-order kinetics where the PFR is more efficient.

Handbook: Environmental Engineering — Steady-State Reactor Parameters table, Ideal CMFR, Zero-Order: $\theta = (C_0 - C_t)/k$ (p. 330–331) • Handbook p. 330–331 — Steady-State Reactor Parameters (Constant Density Systems)

Q9. [D10-Q09 • Numeric • CMFR — second-order reaction, effluent concentration]

A CMFR operates with a second-order decay reaction ($r = -kC^2$) with rate constant $k = 0.02 \text{ L}/(\text{mg}\cdot\text{d})$ and a mean retention time $\theta = 5 \text{ d}$. The influent concentration is $C_0 = 50 \text{ mg/L}$. The effluent concentration C_t is most nearly:

For a CMFR with second-order decay: $C_t = \frac{\sqrt{4kC_0\theta + 1} - 1}{2k\theta}$

Relevant: $C_t = \frac{\sqrt{4kC_0\theta + 1} - 1}{2k\theta}$

Numeric entry.

Answer: 1

$$C_t = \frac{\sqrt{4(0.02)(50)(5)} + 1 - 1}{2(0.02)(5)} = \frac{\sqrt{20} - 1}{0.2} = \frac{4.583 - 1}{0.2} = \frac{3.583}{0.2} = 17.9 \text{ mg/L.}$$

Step by step: $4kC_0\theta = 4(0.02)(50)(5) = 20$; inner radical: $\sqrt{20} = 4.583$; numerator: $4.583 - 1 = 3.583$; denominator: $2(0.02)(5) = 0.2$; $C_t = 3.583/0.2 = 17.9 \text{ mg/L}$.

Option (A) 14.6 mg/L: uses incorrect radical expansion. Option (C) 25.0 mg/L: uses first-order formula $C_0/(1 + k\theta)$ with wrong k units. Option (D) 35.7 mg/L: neglects the subtraction of 1 in the numerator. The second-order CMFR requires solving a quadratic steady-state balance, yielding the formula above.

Handbook: Environmental Engineering — Steady-State Reactor Parameters table, Ideal CMFR, Second-Order: C_t formula (p. 330–331) • Handbook p. 330–331 — Steady-State Reactor Parameters (Constant Density Systems)

Q10. [D10-Q10 • MCQ • CMFR vs PFR — comparison for first-order removal]

A treatment process must achieve 90% removal of a contaminant that degrades via first-order kinetics. Two reactor configurations are considered: an ideal CMFR and an ideal PFR, each with the same first-order rate constant k . Which statement best compares the required hydraulic retention times (θ)?

$$\text{Relevant: } \theta_{\text{CMFR}} = \frac{C_0/C_t - 1}{k} \quad \theta_{\text{PFR}} = \frac{\ln(C_0/C_t)}{k}$$

- (A) $\theta_{\text{CMFR}} < \theta_{\text{PFR}}$ (CMFR is more efficient)
- (B) $\theta_{\text{CMFR}} \approx \theta_{\text{PFR}}$ (both reactors perform equally)
- (C) $\theta_{\text{CMFR}} \approx 3.9 \times \theta_{\text{PFR}}$ (CMFR requires $\sim 4 \times$ longer HRT) ✓
- (D) $\theta_{\text{CMFR}} = 9 \times \theta_{\text{PFR}}$ (CMFR requires exactly $9 \times$ longer HRT)

Answer: (C)

For 90% removal: $C_0/C_t = 10$.

$$\theta_{\text{CMFR}} = \frac{10 - 1}{k} = \frac{9}{k}$$

$$\theta_{\text{PFR}} = \frac{\ln 10}{k} = \frac{2.303}{k}$$

$$\text{Ratio: } \theta_{\text{CMFR}}/\theta_{\text{PFR}} = 9/2.303 = 3.91$$

Option (A): wrong — the CMFR is less efficient, not more, because it operates at the (lowest) effluent concentration throughout the entire volume. Option (B): wrong — they differ significantly for first-order kinetics. Option (D): $\theta_{\text{CMFR}} = 9/k$ and $\theta_{\text{PFR}} = \ln(10)/k \approx 2.3/k$, so the ratio is 3.9, not 9. The PFR advantage exists because it maintains a high concentration (high reaction rate) along the length of the reactor.

Handbook: Environmental Engineering — Steady-State Reactor Parameters table, Ideal CMFR and Ideal Plug Flow, First-Order (p. 330–331) • Handbook p. 330–331 — Steady-State Reactor Parameters (Constant Density Systems)

Q11. [D10-Q11 • MCQ • Engineering stress — axial loading (SI)]

A steel rod with an initial diameter of 20 mm carries an axial tensile load of 40 kN. The engineering stress in the rod is most nearly:

Engineering stress: $= F/A_0$

Relevant: $\sigma = \frac{F}{A_0}$ $A_0 = \frac{\pi d^2}{4}$

- (A) 63.7 MPa
- (B) 100 MPa
- (C) **127 MPa ✓**
- (D) 159 MPa

Answer: (C)

Initial cross-sectional area:

$$A_0 = \frac{\pi(0.020)^2}{4} = \frac{\pi \times 4 \times 10^{-4}}{4} = 3.142 \times 10^{-4} \text{ m}^2$$

Engineering stress:

$$\sigma = \frac{F}{A_0} = \frac{40,000 \text{ N}}{3.142 \times 10^{-4} \text{ m}^2} = 127.3 \times 10^6 \text{ Pa} \approx 127 \text{ MPa}$$

Option (A) 63.7 MPa: uses diameter instead of radius in computing area ($A = \pi D^2/4$ not πD^2). Option (B) 100 MPa: error in unit conversion. Option (D) 159 MPa: uses $r = 20$ mm (full diameter as radius). Engineering stress always uses the original (undeformed) cross-sectional area A_0 .

Handbook: Materials Science/Structure of Matter — Engineering stress: $= F/A_0$, where A_0 = initial cross-sectional area (p. 121) • Handbook p. 121 — Materials Science/Structure of Matter: Stress and Strain

Q12. [D10-Q12 • MCQ • Young's modulus — Hooke's Law (SI)]

A specimen of aluminum alloy is subjected to a uniaxial tensile stress of 140 MPa, producing an elastic strain of 0.002 (dimensionless). The elastic modulus (Young's modulus) of the material is most nearly:

Hooke's Law: $\sigma = E\varepsilon$

Relevant: $E = \frac{\sigma}{\varepsilon}$

- (A) 28 GPa
- (B) 56 GPa
- (C) **70 GPa** ✓
- (D) 140 GPa

Answer: (C)

$$E = \frac{\sigma}{\varepsilon} = \frac{140 \times 10^6 \text{ Pa}}{0.002} = 70 \times 10^9 \text{ Pa} = 70 \text{ GPa}$$

This is consistent with the elastic modulus of aluminum (~70 GPa).

Option (A) 28 GPa: divides by 5 instead of 0.002 (misplaces decimal). Option (B) 56 GPa: arithmetic error. Option (D) 140 GPa: uses $E = \sigma/\varepsilon$ but forgets to apply ε (divides by 1 instead of 0.002). Hooke's Law applies only in the elastic (linear) region of the stress-strain curve.

Handbook: Materials Science/Structure of Matter — Hooke's Law: $\sigma = E\varepsilon$, where E = elastic modulus (also called modulus, modulus of elasticity, Young's modulus) (p. 121) • Handbook p. 121 — Materials Science/Structure of Matter: Elastic modulus (Young's modulus)

Q13. [D10-Q13 • MCQ • Hooke's Law — elastic strain from applied stress (USCS)]

A structural steel member carries an axial stress of 29,000 psi. The steel has an elastic modulus $E = 29 \times 10^6$ psi (given). The elastic strain is most nearly:

Hooke's Law: $\sigma = E\varepsilon$

Relevant: $\varepsilon = \frac{\sigma}{E}$

- (A) 0.0005
- (B) **0.0010** ✓
- (C) 0.0015
- (D) 0.0020

Answer: (B)

$$\varepsilon = \frac{\sigma}{E} = \frac{29,000 \text{ psi}}{29 \times 10^6 \text{ psi}} = 1.0 \times 10^{-3} = 0.0010$$

Option (A) 0.0005: divides by 58×10^6 (doubles E). Option (C) 0.0015: arithmetic error. Option (D) 0.0020: uses $E = 14.5 \times 10^6$ psi (half of correct value). At $\varepsilon = 0.001$, this steel is comfortably in the elastic range (yield strain for structural steel ≈ 0.001 – 0.002).

Handbook: Materials Science/Structure of Matter — Hooke's Law: $\sigma = E\varepsilon$ (p. 121). $E = 29 \times 10^6$ psi for steel is provided in the stem. • Handbook p. 121 — Hooke's Law $\sigma = E\varepsilon$; E value provided in stem

Q14. [D10-Q14 • MCQ • True stress vs engineering stress]

A tensile test specimen has an initial cross-sectional area $A_0 = 200 \text{ mm}^2$ and is loaded with $F = 30,000 \text{ N}$. At a point during plastic deformation, the actual (current) cross-sectional area has necked to $A = 170 \text{ mm}^2$. The true stress at this point is most nearly:

True stress: $T = F/A$ — Engineering stress: $= F/A_0$

Relevant: $\sigma_T = \frac{F}{A}$ $\sigma = \frac{F}{A_0}$

- (A) 150.0 MPa (engineering stress — uses initial area)
- (B) 164.7 MPa
- (C) **176.5 MPa (true stress — uses actual area) ✓**
- (D) 188.2 MPa

Answer: (C)

True stress uses the actual (current) cross-sectional area:

$$\sigma_T = \frac{F}{A} = \frac{30,000 \text{ N}}{170 \times 10^{-6} \text{ m}^2} = 176.5 \times 10^6 \text{ Pa} = 176.5 \text{ MPa}$$

Engineering stress (for comparison):

$$\sigma = \frac{F}{A_0} = \frac{30,000}{200 \times 10^{-6}} = 150.0 \text{ MPa}$$

Option (A) 150 MPa: this is the engineering stress — uses undeformed A_0 . Option (B) 164.7 MPa: uses an intermediate area estimate. Option (D) 188.2 MPa: uses $A = 160 \text{ mm}^2$ (too small). True stress exceeds engineering stress after yielding because the area decreases. Before yielding they are essentially equal.

Handbook: Materials Science/Structure of Matter — True stress: $T = F/A$ (actual area); Engineering stress: $= F/A_0$ (initial area). 'True stress is load divided by actual cross-sectional area whereas engineering stress is load divided by the initial area.' (p. 121) • Handbook p. 121 — Materials Science/Structure of Matter: True stress and Engineering stress

Q15. [D10-Q15 • MCQ • Thermal deformation — axial member (SI)]

A structural steel member is 4.0 m long. It is heated from an initial temperature $T_0 = 20^\circ\text{C}$ to a final temperature $T = 80^\circ\text{C}$. The thermal expansion coefficient of steel is $\alpha = 12 \times 10^{-6} / ^\circ\text{C}$. The thermal deformation t is most nearly:

Thermal Deformations: $t = \alpha L(T - T_0)$

Relevant: $\delta_t = \alpha L(T - T_0)$

- (A) 1.4 mm
- (B) **2.9 mm** ✓
- (C) 3.6 mm
- (D) 5.0 mm

Answer: (B)

$$\begin{aligned}\delta_t &= \alpha L(T - T_0) = (12 \times 10^{-6} / ^\circ\text{C} / ^\circ\text{C} / ^\circ\text{C})(4.0 \text{ m})(80 - 20) \\ &= 12 \times 10^{-6} \times 4.0 \times 60 = 2,880 \times 10^{-6} \text{ m} = 2.88 \text{ mm} \approx 2.9 \text{ mm}\end{aligned}$$

Option (A) 1.4 mm: uses $T = 30^\circ\text{C}$ (misreads $T = T - T_0$ as $T/2$). Option (C) 3.6 mm: uses $T = 75^\circ\text{C}$ or $L = 5 \text{ m}$. Option (D) 5.0 mm: uses $\alpha = 21 \times 10^{-6}$ (incorrect for steel). The formula is in the Mechanics of Materials section; t is positive for expansion ($T > T_0$).

Handbook: Mechanics of Materials — Thermal Deformations: $t = \alpha L(T - T_0)$, where t = deformation caused by a change in temperature, α = temperature coefficient of expansion (p. 131) • Handbook p. 131 — Mechanics of Materials: Thermal Deformations

Q16. [D10-Q16 • MCQ • Thermal expansion coefficient — strain from temperature change]

The thermal expansion coefficient of an aluminum alloy is $\alpha = 23 \times 10^{-6} / ^\circ\text{C}$. A component is heated by $T = 50^\circ\text{C}$. The resulting thermal engineering strain ε is most nearly:

Thermal Properties: $\varepsilon = \alpha T$

Relevant: $\varepsilon = \alpha \Delta T$

- (A) 0.00092
- (B) **0.00115** ✓
- (C) 0.00138
- (D) 0.00160

Answer: (B)

$$\varepsilon = \alpha \Delta T = (23 \times 10^{-6} / ^\circ\text{C} / ^\circ\text{C} / ^\circ\text{C} / ^\circ\text{C})(50)$$

Option (A) 0.00092: uses $\alpha = 18.4 \times 10^{-6}$ (value for a different material). Option (C) 0.00138: uses $T = 60^\circ\text{C}$. Option (D) 0.00160: uses $\alpha = 32 \times 10^{-6}$. This formula gives the free thermal strain — the strain that would occur in an unconstrained member. If the member is restrained, a thermal stress $= E\varepsilon = E\alpha T$ develops.

Handbook: Materials Science/Structure of Matter — Thermal Properties: 'The thermal expansion coefficient is the ratio of engineering strain to the change in temperature.' $\varepsilon = \alpha T$ (p. 127) • Handbook p. 127 — Materials Science: Thermal Properties (thermal expansion coefficient)

Q17. [D10-Q17 • MCQ • Fire hydrant — rated capacity at 20 psi residual]

During a fire hydrant flow test, the static pressure is $PS = 80$ psi and the residual pressure is $PR = 45$ psi at a total test flow of $QF = 1,200$ gpm. The rated capacity QR at 20 psi residual is most nearly:

Fire Hydrant Rated Capacity (NFPA 291): $QR = QF \times (HR/HF)^{0.54}$ where $HR = PS - 20$ and $HF = PS - PR$

Relevant: $Q_R = Q_F \times \left(\frac{H_R}{H_F} \right)^{0.54}$

- (A) 1,200 gpm
- (B) 1,400 gpm
- (C) **1,600 gpm ✓**
- (D) 1,800 gpm

Answer: (C)

$$H_R = P_S - 20 = 80 - 20 = 60 \text{ psi}$$

$$H_F = P_S - P_R = 80 - 45 = 35 \text{ psi}$$

$$Q_R = 1,200 \times \left(\frac{60}{35} \right)^{0.54} = 1,200 \times (1.714)^{0.54}$$

$$(1.714)^{0.54}: \ln(1.714) = 0.538; 0.54 \times 0.538 = 0.291; e^{0.291} = 1.338.$$

$$Q_R = 1,200 \times 1.338 = 1,605 \text{ gpm} \approx 1,600 \text{ gpm}$$

Option (A) 1,200 gpm: the test flow — no extrapolation applied. Option (B) 1,400 gpm: underestimates the exponent or drops the 0.54. Option (D) 1,800 gpm: uses exponent 1.0 (linear extrapolation) instead of 0.54.

Handbook: Civil Engineering — Formula for Calculating Rated Capacity at 20 psi from Fire Hydrant: $QR = QF \times (HR/HF)^{0.54}$, Source: NFPA Standard 291 (p. 304) • Handbook p. 304 — Civil Engineering: Fire Hydrant Rated Capacity (NFPA 291)

Q18. [D10-Q18 • MCQ • Fire hydrant — discharge to atmosphere]

A fire hydrant with a smooth, well-rounded outlet ($C_d = 0.90$) has an outlet diameter of 2.5 in. A pitot gauge reading at the nozzle indicates $P = 36$ psi. The flow rate discharged to the atmosphere is most nearly:

Fire Hydrant Discharging to Atmosphere (NFPA 291): $Q = 29.8 D^2 C_d P^{1/2}$

Relevant: $Q = 29.8 D^2 C_d P^{1/2}$

- (A) 670 gpm
- (B) 840 gpm
- (C) **1,000 gpm** ✓
- (D) 1,340 gpm

Answer: (C)

$$\begin{aligned}
 Q &= 29.8 \times (2.5)^2 \times 0.90 \times \sqrt{36} \\
 &= 29.8 \times 6.25 \times 0.90 \times 6.0 \\
 &= 29.8 \times 6.25 = 186.25 \\
 &\quad \times 0.90 = 167.6 \\
 &\quad \times 6.0 = 1,006 \text{ gpm} \approx 1,000 \text{ gpm}
 \end{aligned}$$

Option (A) 670 gpm: uses $C_d = 0.60$ (wrong coefficient) or misses the D^2 squaring. Option (B) 840 gpm: uses $C_d = 0.75$ (intermediate coefficient for a square-edged outlet). Option (D) 1,340 gpm: omits C_d (treats $C_d = 1.0$). The smooth, well-rounded outlet has $C_d = 0.90$; square-edged = 0.80; projecting into barrel = 0.70 (per NFPA 291 table in Handbook).

Handbook: Civil Engineering — Fire Hydrant Discharging to Atmosphere: $Q = 29.8 D^2 C_d P^{1/2}$, $C_d = 0.90$ for smooth, well-rounded outlet (p. 304–305, NFPA 291) • Handbook p. 304–305 — Civil Engineering: Fire Hydrant Discharging to Atmosphere (NFPA 291)